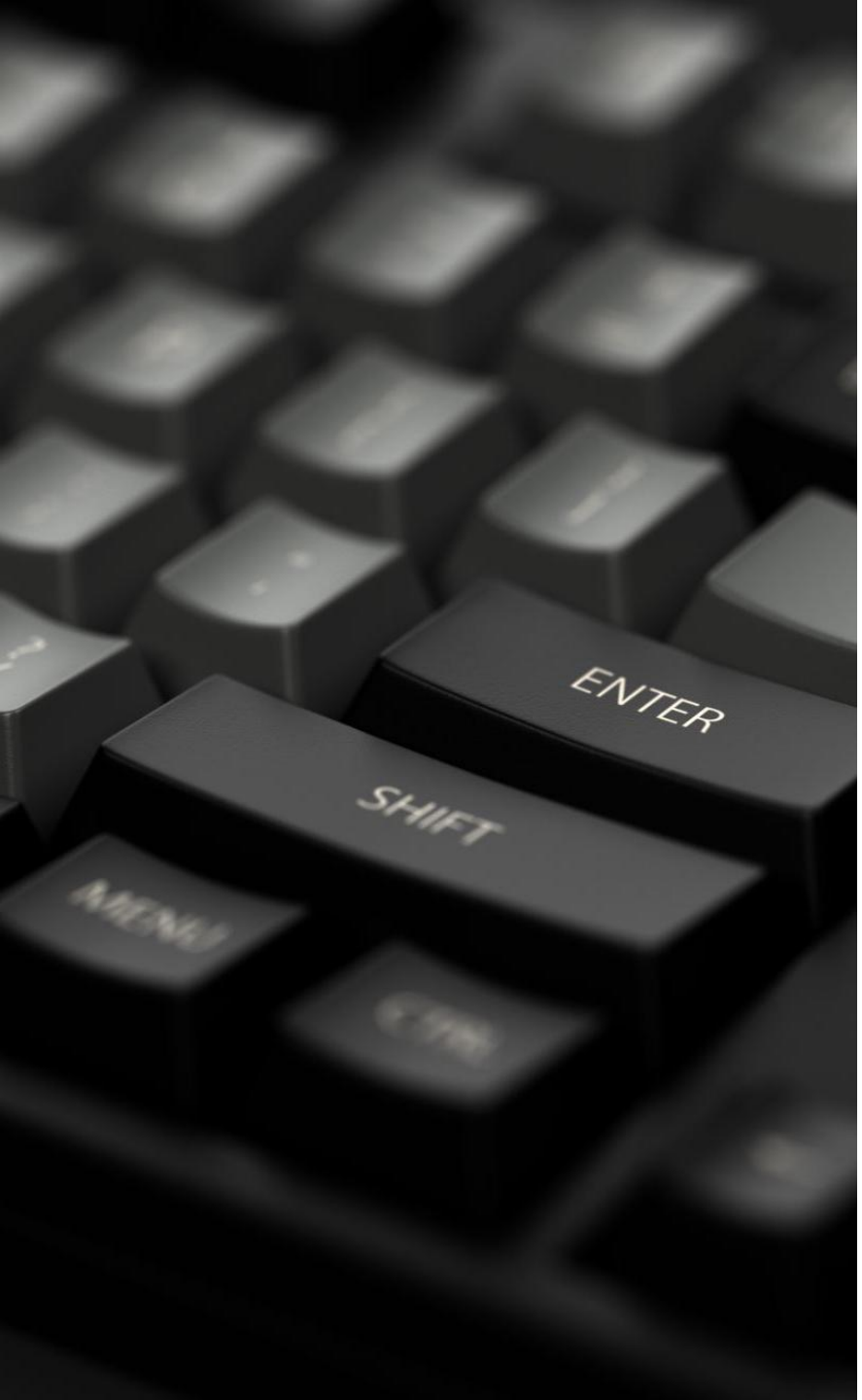
A gamified introduction to Python Programming

Lecture 13
Importing and Project Management

Matthieu DE MARI – Singapore University of Technology and Design



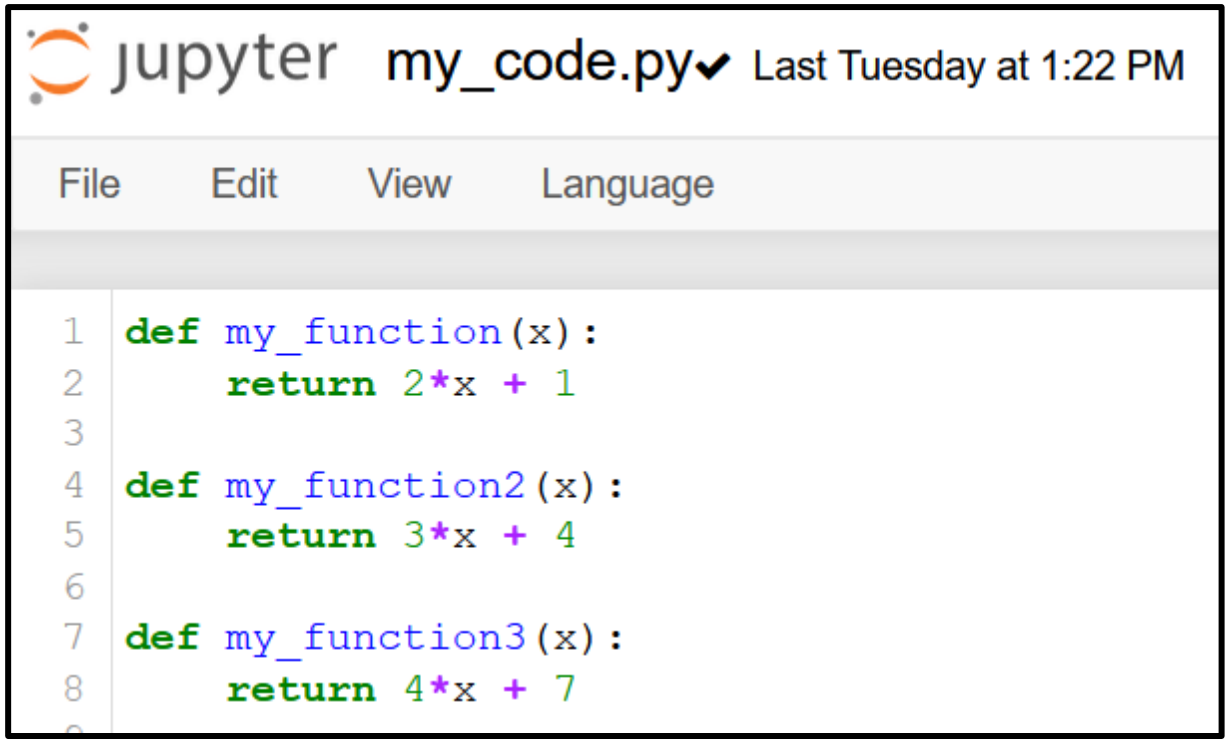
Outline (Chapter 13)

- What is the **import procedure** doing exactly?
- How to organize your different codes for better **project organizing**?
- **Our first mini-project**.

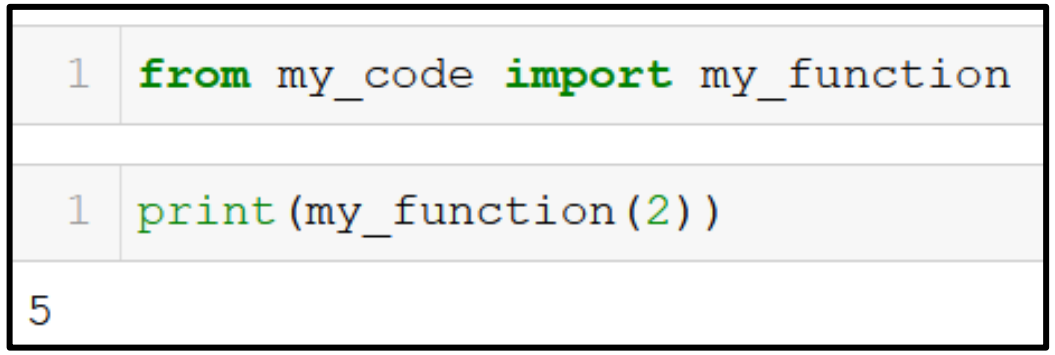
The import procedure

- The import procedure is used to **import functions defined in external .py files.**
- To demonstrate, we have defined a **my_code.py** file with three functions.
- We can then import one of these functions in our Notebook, by using the **from … import …** command.

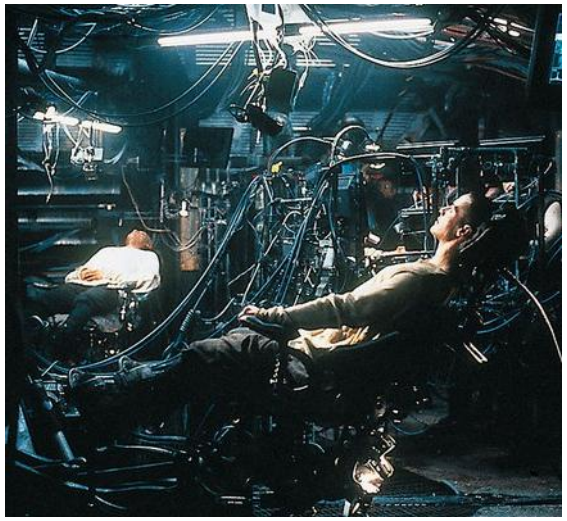
my_code.py file



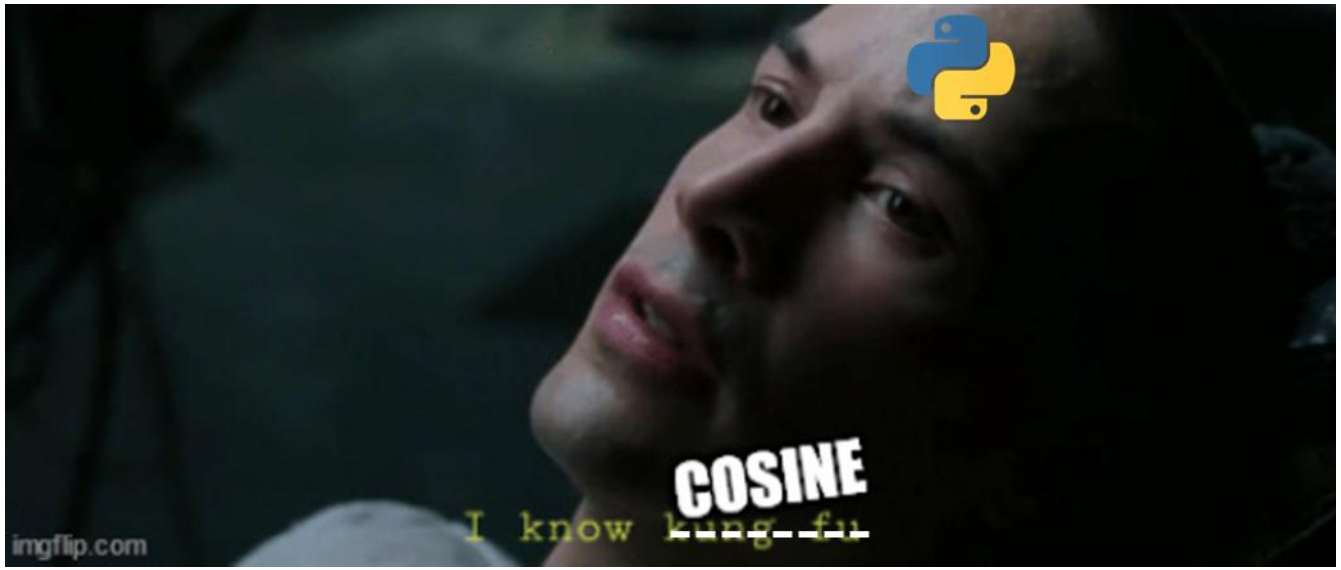
In our notebook



The import procedure



Operator:
Cosine function coming right up!




```
1 print(cos(0))
```

Importing as

If needed, we can **import** and **rename** a function by using the **as** keyword after an **import**.

- The whole command then reads **from … import … as …**

- **Note:** if you rename the function, its calling name changes to the alias you specified.

my_code.py file

jupyter my_code.py ✔ Last Tuesday at 1:22 PM

File Edit View Language

```
1  def my_function(x):
2      return 2*x + 1
3
4  def my_function2(x):
5      return 3*x + 4
6
7  def my_function3(x):
8      return 4*x + 7
```

In our notebook

```
1  from my_code import my_function2 as custom_name
```

```
1  print(my_function2(2))
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
<ipython-input-4-20b74e0a7273> in <module>
----> 1 print(my_function2(2))

NameError: name 'my_function2' is not defined
```

```
1  print(custom_name(2))
```

```
10
```

Importing several functions

If needed, you can import **multiple functions at once**.

- Simply use **commas (,) symbols** to separate the different functions names.
- Or, you can also **import all functions** at once, by simply entering *.

(This is considered malpractice, however, you should just import what you need!)

my_code.py file

jupyter my_code.py ✔ Last Tuesday at 1:22 PM

File Edit View Language

```
1  def my_function(x):
2      return 2*x + 1
3
4  def my_function2(x):
5      return 3*x + 4
6
7  def my_function3(x):
8      return 4*x + 7
```

In our notebook

```
1  from my_code import my_function, my_function2
```

```
1  from my_code import *
```

```
1  print(my_function3(2))
```

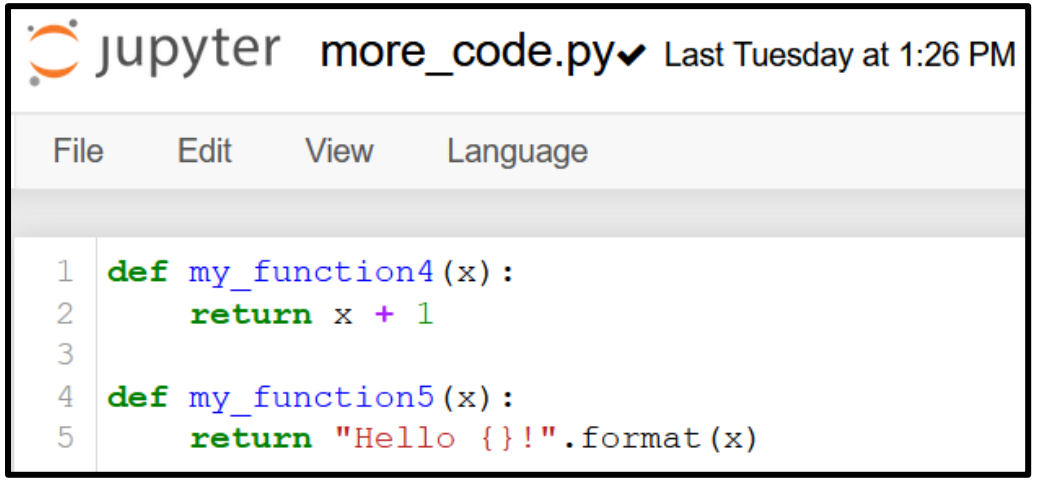
```
15
```

Importing an entire module

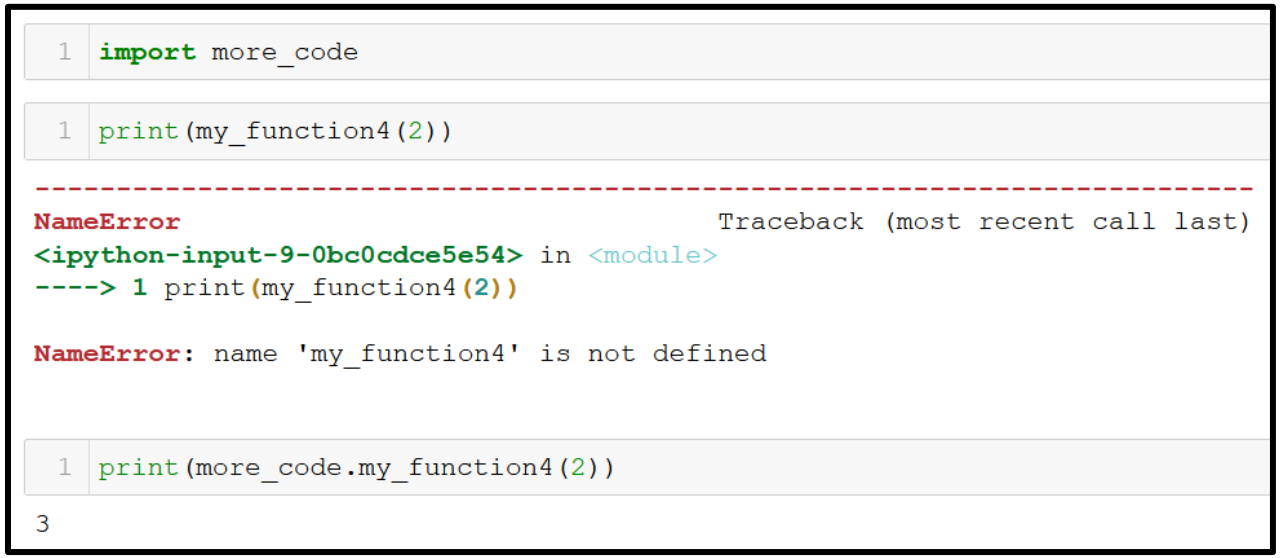
You can also import a whole file or library as a **module.**

- To do so, simply start with **import** instead of **from**.
- By doing so, you import all the functions, but **they have to be called using the module name and the dot operator** (.).
- This can make the code more readable, but is also a bit more inconvenient (your choice!).

more_code.py file



In our notebook



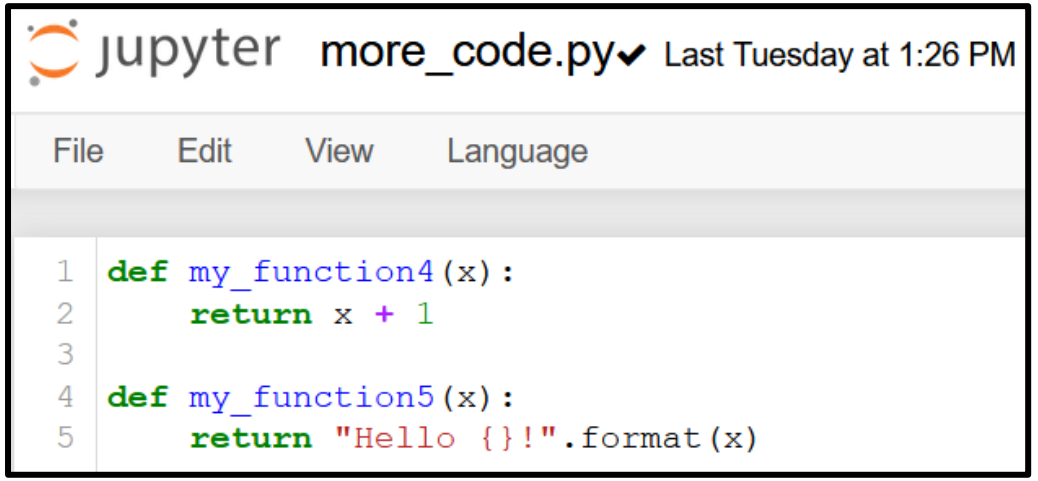
Importing an entire module

You can also use an alias for the module using the **as** keyword, as before.

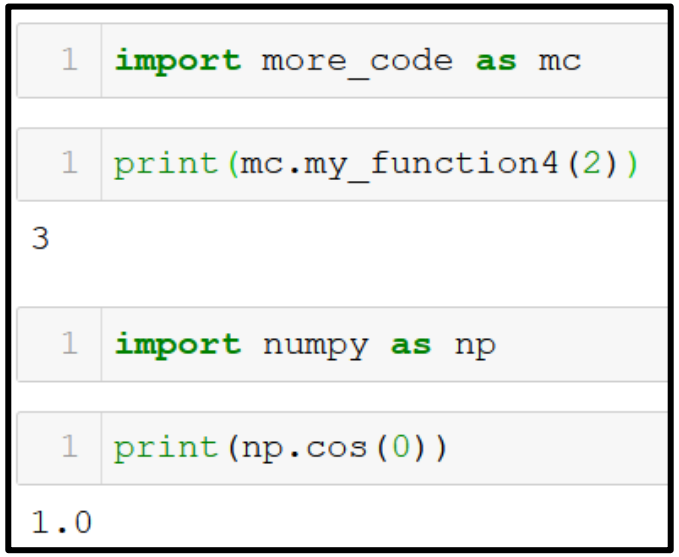
- Typically, we often do it with the Numpy library!

import numpy as np

more_code.py file



In our notebook



Folders to import from

Observation: Python imported functions from my_code.py, which was located in the same folder as my Notebook.

- **But there was no numpy.py file in this location.**
- **What happened?**

- Python looks for files in your current folder first,
- And then looks in your Python installation directory.

- When you installed the Numpy package with **pip** on Week 1, you downloaded some **numpy files** and stored them in your Python installation folder!

Folders to import from

- Python looks for files in your current folder first,
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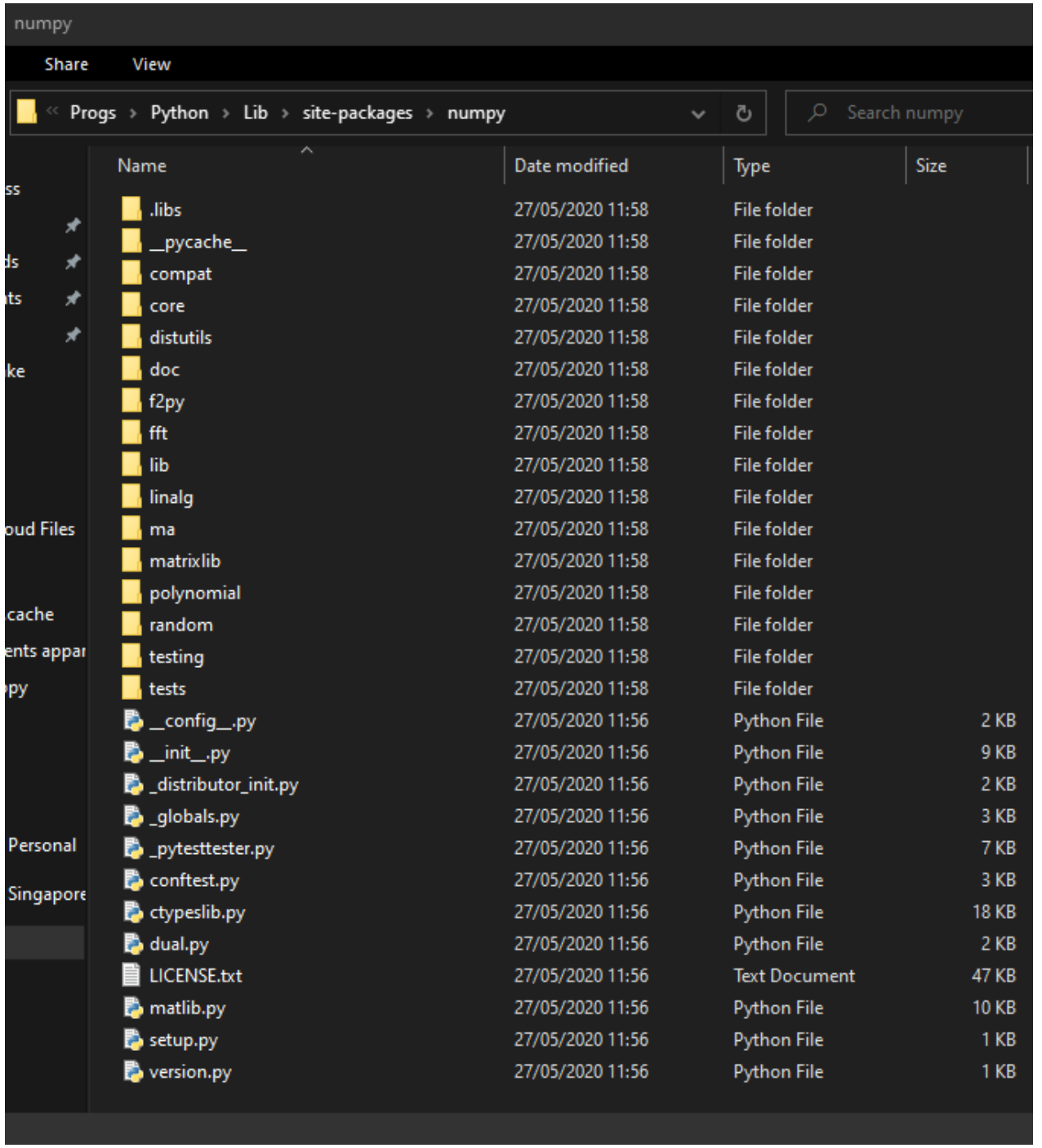
- When you installed the Numpy package with **pip** on Week 1, you downloaded some **numpy** files and stored them in your Python installation folder!



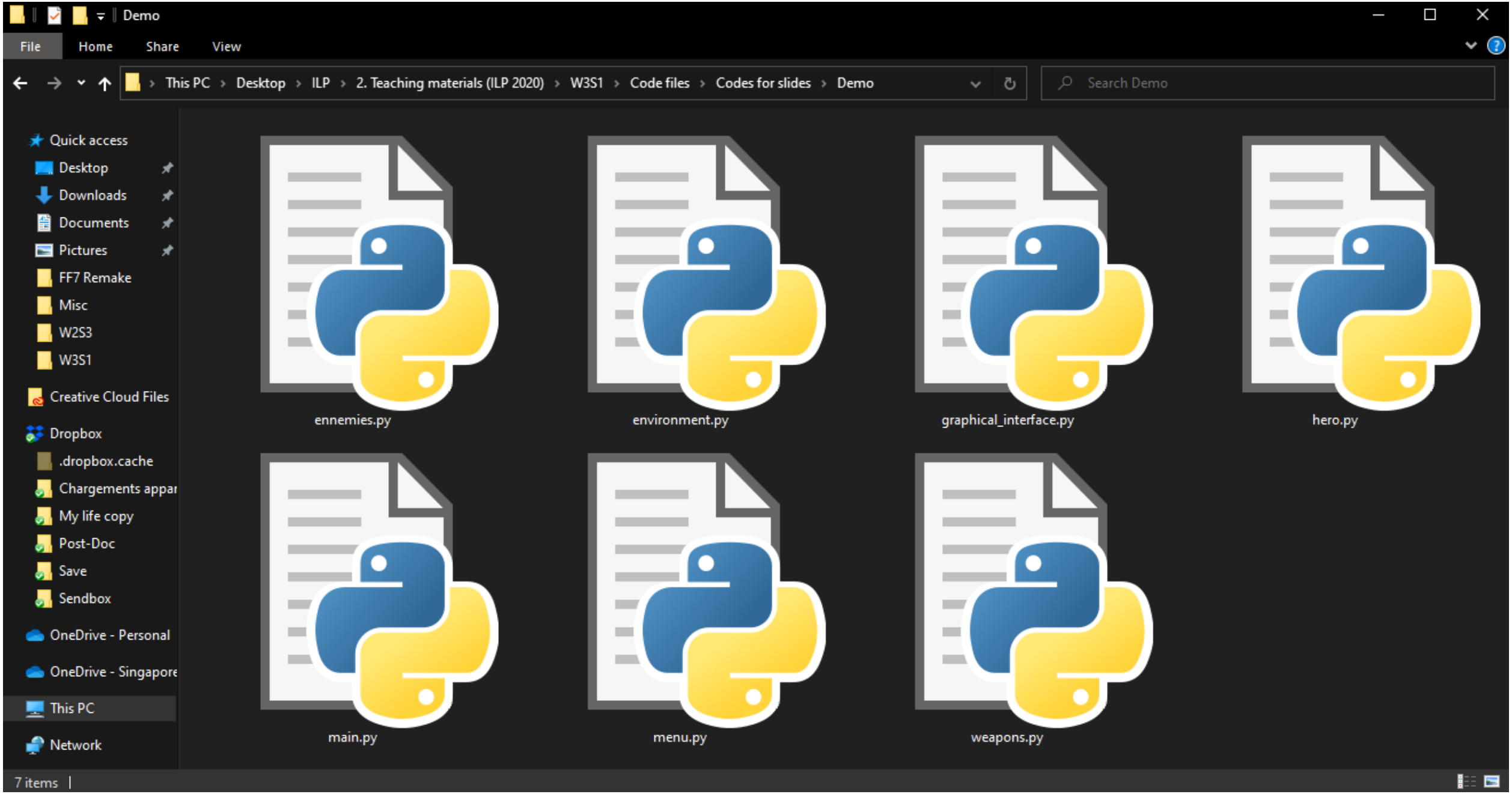
Project organizing

Question: How to organize your code/projects, when you start to have many functions working together?

- In many projects, several developers will often collaborate on a single project.
- And each one of them will end up working on a single aspect of the project.

E.g. in a video game:

- One developer can design the main character,
- Another one will develop enemies,
- Another one will develop items and weapons,
- Another one will design the map/environment in which heroes and enemies evolve,
- Etc.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Downloads
Documents
Pictures
FF7 Remake
Misc
W2S3
W3S1
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Dropbox
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Sendbox
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OneDrive - Singapore
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environment.py
graphical_interface.py
hero.py
main.py
menu.py
weapons.py
7 items

Project organizing

In practice,

- Each developer will then work on his/her own .py file, taking care of his/her specific subtask.
- Later on, other developers might **import** functions from files created by other people.
- Eventually assemble all pieces in a **main.py** file!

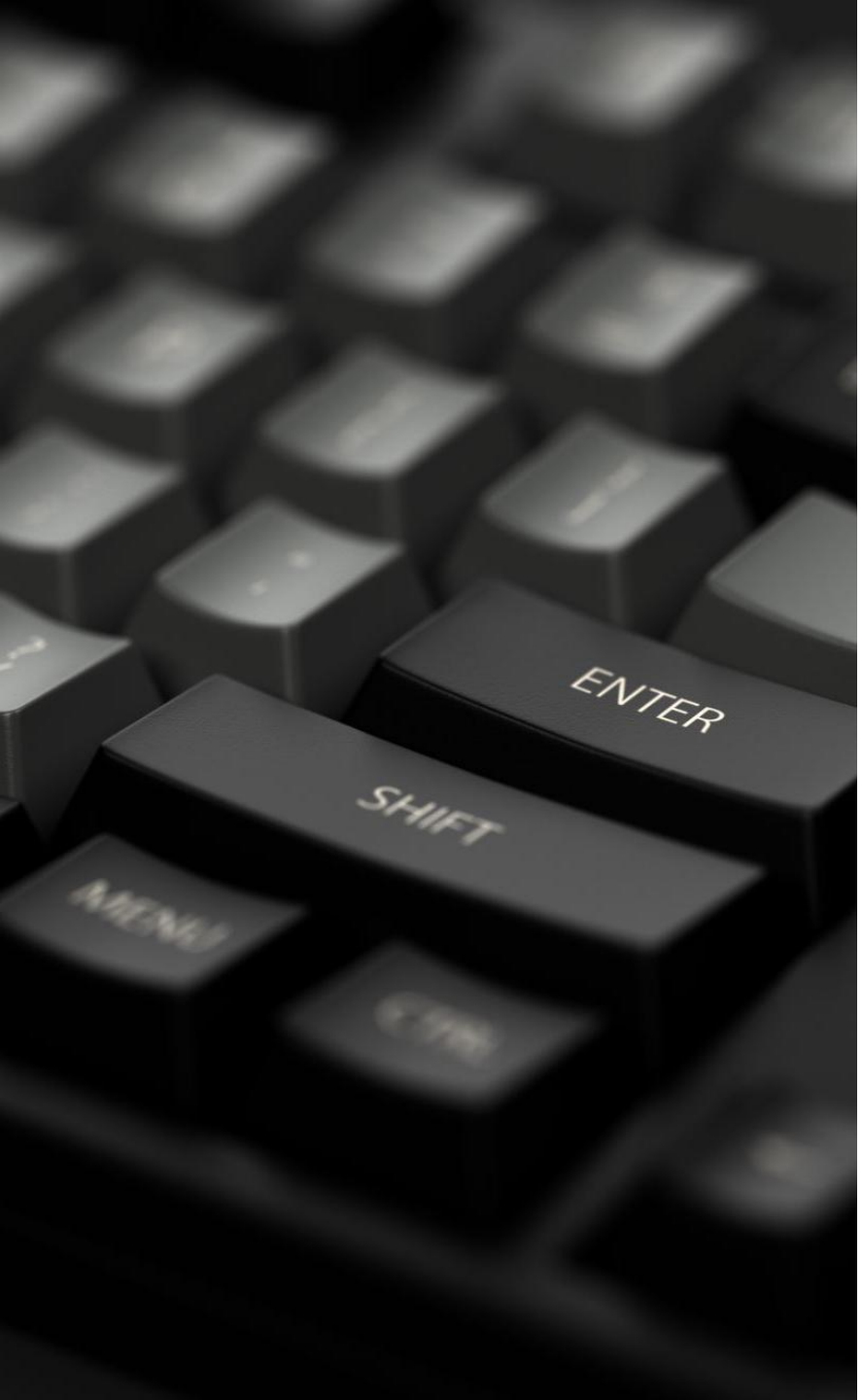
- This is something very common in programming projects.

- **Important:** If your code will be reused by other people, even though they have not coded it, you need to build the habit of documenting your functions! *(Otherwise, they will need to read your code to understand it and they will hate you for it)*

Matt's Great advice

Matt's Great Advice: Good import practices and project organizing.

1. As with the variables and functions, it is a good idea to **make your file names explicit**.
2. Have **a file for each sub-concept**, and **a single main file that assembles them all** at the end.
3. It is often better to **import only what is needed** (**from** … **import** …) rather than importing everything which is overkill. (**from** … **import** *, **import** …).
4. Comment your code! (Don't be THAT guy).



Conclusion (Chapter 12)

- What is the **import procedure** doing exactly?
- How to organize your different codes for better **project organizing**?
- **Our first mini-project**.

Let us move on to our first Mini-project now!

Wait for me to explain and show you first.